

Vineet Dairashri

Master's Student in Data Science | Machine Learning with Tabular Data

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PROFILE SUMMARY

Master's student in Data Science at TU Dortmund with hands-on experience building end-to-end machine learning pipelines in Python starting from feature engineering on tabular and sensor data to statistical modeling, probabilistic inference, and rigorous, reproducible evaluation. Experienced with supervised learning, Bayesian and simulation-based inference, and calibration diagnostics that make model outputs transparent and trustworthy. Comfortable working with large, noisy, real-world datasets and translating recurring patterns into interpretable, explainable decision logic. Motivated by applying statistical and machine learning methods to deliver scalable, trustworthy analysis for real-world engineering and vehicle data.

SKILLS

Programming Languages

Python, R, SQL, C/C++ (basics)

Machine Learning & Statistical Modeling

PyTorch, TensorFlow, Scikit-learn, Transformer, XGBoost, TabICLv2, k-means/DBSCAN, Representation Learning, Feature Engineering, Statistical Theory

Data Processing & Visualization

Pandas, NumPy, Matplotlib, Tableau

Explainable & Robust ML

Model Evaluation & Benchmarking, Ablation Studies, Calibration Diagnostics, Interpretable Modeling, Clustering & Pattern Analysis, Robust Learning under Noisy/Imperfect Data

Tools & Platforms

Git/Github, Docker, AWS, Azure. Microsoft Office

Language Skills

English (C1), German (A2, learning toward B1)

EXPERIENCE

Research Assistant

01/2025 - Present University of Duisburg-Essen, Germany

- Build reproducible analytical and computational workflows under version control, with structured evaluation and clear documentation aligned with rigorous research practice.
- Extend statistical learning methods by implementing robust split-rule logic, improving model reliability under noisy, skewed, or imperfect data.
- Design analytical tasks and assignments, requiring clean code, methodological rigor, and clear written and verbal communication of technical results.

Data Analyst

02/2023 - 09/2023 OneknotOne Technologies, India

- Built a recommendation system in Python using a KD Tree-based similarity search over 50,000+ user-item interactions, applying tree/graph data structures to scale neighbour queries.
- Improved user engagement by 15% and reduced computational complexity from $O(n^2)$ to $O(n \log n)$ through algorithmic redesign.
- Deployed and benchmarked the model on AWS, reducing processing time by 20% and establishing a reproducible inference pipeline.

RESEARCH PROJECTS

- 04/2025 - 09/2025 **Master's Case Study: Classification with Deep Learning** **Grade: 1.0**
 - Designed and trained an end-to-end deep learning pipeline in PyTorch for a large-scale, fine-grained classification task, with a modular architecture suitable for adapting to sequence and graph-structured inputs.
 - Implemented, benchmarked, and ensembled CNN-based architectures with hyperparameter tuning and ablation studies; achieved 0.880 AUC-ROC, demonstrating rigorous evaluation, robustness analysis, and iterative model improvement.
- 04/2025 - 08/2025 **Simulation-Based Inference of Cosmological Parameters** **Grade: 2.0**
 - Built a simulation-based inference pipeline using BayesFlow to recover physical parameters from noisy power spectra, demonstrating probabilistic reasoning useful for uncertainty-aware path prediction.
 - Combined LSTNet sequence networks with normalising flows (CouplingFlow) for amortised posterior estimation and validated calibration with SBC, contraction, and z-score diagnostics.
- 04/2025 - 08/2025 **Wellbore Geology Prediction (Kaggle competition)**
 - Built ML pipeline to predict target values from tabular well-log sensor data, engineering features from raw numeric/time-series columns per well.
 - Combined multiple tabular models (LightGBM, CatBoost, XGBoost) into a stacked ensemble, later adding a tabular foundation model (TabICL) as an extra signal, to boost prediction accuracy on the tabular dataset.
 - Iteratively improved the tabular pipeline across several versions to 10.5 RMSE, steadily boosting leaderboard performance.

EDUCATION

- 09/2023 - Present **Master of Data Science**, Technische Universität Dortmund **Current GPA: 2.5**
Advanced Machine Learning, Deep Learning, Machine Learning for Big Data, Statistical Theory, Applied Bayesian Methods, Causality, Decision Tree & Optimization, Industrial Data Science, Fintech & AI in Finance
Erasmus: Fintech and AI in Finance at University of Naples Federico II
- 07/2019 - 06/2023 **Bachelor of Engineering in Information Technology** **GPA: 1.7**
Sardar Vallabhbhai Patel Institute of Technology, India
Data Structures & Algorithms, SQL, Programming with C, Computer Vision, Software Project Management, Agile Development, OOP with Java
Bachelor Thesis: Graph-based Recommendation System using KD Tree